University of Engineering & Technology Peshawar

Computer Programming (C++) Lab

Assignment no 1

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BFNWELE0670

SUBMITTED TO ENGR RIZWAN ULLAH

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Part 1:Single C++ Statements

1. Write a single C++ statement to accomplish each of the following:
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3. Display the message "Welcome to Object Oriented Programming in C++!" on the screen.
4. Read a value from the user and store it in an integer variable named age.
5. If the variable temperature is greater than 100, print "Overheating detected!".
6. Compute the area of a circle with radius r and store it in a variable named area (Use the formula Πr^{2}, assume $\Pi = 3.14159$).
7. Print the values of the variables x, y, and z separated by a comma.
8. Assign the sum of voltage and current to a variable named total.
9. Display "Error: Division by zero" only if the variable denominator equals zero.
10. Increment the variable count by 1 and then print its new value.

Part 2: Complex Logic & Calculations

11. Read three floating-point numbers from the keyboard and store them in variables a, b, and c.
12. Print the total resistance when three resistors are connected in series (take the values of the resistances from the user).
13. Print the total resistance when three resistors are connected in parallel (take the values of the resistances from the user).
14. Print the value of the voltage across R2 (using the voltage division rule) when two resistors (R1 and R2) are connected in series. (Select the values for R1, R2, and the voltage yourself).
15. Print the value of the current through R2 (using the current division rule) when two resistors (R1 and R2) are connected in parallel. (Select the values for R1, R2, and the voltage yourself) .
16. Calculate the Semester GPA of your previous semester:
17. a. Enter all your subjects.
18. b. Enter the credit hours.
19. c. Calculate the semester GPA

BHNWELE0670

DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITY OF ENGINEERING & TECHNOLOGY, PESHAWAR
NOWSHERA CAMPUS

Computer Programming — Assignment No. 1

Bachelor of Electrical Engineering (2nd Semester) | Spring 2026

Submission Date: Friday, 23/04/2026

Write a single C++ statement to accomplish each of the following:

PART A — Single C++ Statements (i – x)

i. Declare the variables voltage, current, and resistance to be of type double.

```
double voltage, current, resistance;
```

ii. Display the message "Welcome to Object Oriented Programming in C++!" on the screen.

```
cout << "Welcome to Object Oriented Programming in C++!" << endl;
```

iii. Read a value from the user and store it in an integer variable named age.

```
int age;  cin >> age;
```

iv. If the variable temperature is greater than 100, print "Overheating detected!".

```
if (temperature > 100)  cout << "Overheating detected!" << endl;
```

v. Compute the area of a circle with radius r and store it in a variable named area. (pi = 3.14159)

```
double area = 3.14159 * r * r;
```

vi. Print the values of the variables x, y, and z separated by a comma.

```
cout << x << ", " << y << ", " << z << endl;
```

vii. Assign the sum of voltage and current to a variable named total.

```
double total = voltage + current;
```

viii. Display "Error: Division by zero" only if the variable denominator equals zero.

```
if (denominator == 0)  cout << "Error: Division by zero" << endl;
```

ix. Increment the variable count by 1 and then print its new value.

```
cout << ++count << endl;
```

x. Read three floating-point numbers from the keyboard and store them in variables a, b, and c.

```
float a, b, c;  cin >> a >> b >> c;
```

PART B — Resistor Circuits (xi – xiv)

xi. Print total resistance — three resistors connected in series.

```
#include <iostream>
using namespace std;
int main() {
    double R1, R2, R3;
    cout << "Enter R1, R2, R3 (ohms): ";
    cin >> R1 >> R2 >> R3;
```

```
    double Rtotal = R1 + R2 + R3;
    cout << "Total Resistance (Series) = " << Rtotal << " ohms" << endl;
    return 0;
}
```

Explanation: In a series circuit, resistances add directly: R_total = R1 + R2 + R3

xii. Print total resistance — three resistors connected in parallel.

```
#include <iostream>
using namespace std;
int main() {
    double R1, R2, R3;
    cout << "Enter R1, R2, R3 (ohms): ";
    cin >> R1 >> R2 >> R3;
    double Rtotal = 1.0 / (1.0/R1 + 1.0/R2 + 1.0/R3);
    cout << "Total Resistance (Parallel) = " << Rtotal << " ohms" << endl;
    return 0;
}
```

Explanation: In a parallel circuit: 1/R_total = 1/R1 + 1/R2 + 1/R3

xiii. Voltage across R2 using the Voltage Division Rule (R1 = 200 ohm, R2 = 300 ohm, Vs = 10 V)

```
#include <iostream>
using namespace std;
int main() {
    double R1 = 200.0, R2 = 300.0, Vs = 10.0;
    double V2 = Vs * (R2 / (R1 + R2));
    cout << "Voltage across R2 = " << V2 << " V" << endl;
    return 0;
}
// Output: Voltage across R2 = 6 V
```

Explanation: V2 = Vs x R2 / (R1 + R2) = 10 x 300 / 500 = 6 V

xiv. Current through R2 using the Current Division Rule (R1 = 100 ohm, R2 = 200 ohm, Vs = 12 V)

```
#include <iostream>
using namespace std;
int main() {
    double R1 = 100.0, R2 = 200.0, Vs = 12.0;
    double Rp   = (R1 * R2) / (R1 + R2);   // parallel equivalent
    double Itot = Vs / Rp;                  // total current from source
    double I2   = Itot * (R1 / (R1 + R2));  // current divider formula
    cout << "Current through R2 = " << I2 << " A" << endl;
    return 0;
}
// Output: Current through R2 = 0.08 A
```

Explanation: I2 = I_total x R1 / (R1 + R2) — current splits inversely proportional to resistance.

PART C — Semester GPA Calculator (xv)

xv. Calculate the Semester GPA — enter all subjects, credit hours, and calculate GPA.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    int n;
    cout << "Enter number of subjects: ";
    cin >> n;
```

```
    double totalPoints = 0, totalCredits = 0;

    for (int i = 0; i < n; i++) {
        string subject;
        double credits, gradePoint;

        cout << "\nEnter subject name: ";
        cin >> subject;

        cout << "Enter credit hours: ";
        cin >> credits;

        cout << "Enter grade point (A=4, B=3, C=2, D=1, F=0): ";
        cin >> gradePoint;

        totalPoints += credits * gradePoint;
        totalCredits += credits;
    }

    if (totalCredits == 0) {
        cout << "\nInvalid input! Credits cannot be zero." << endl;
        return 0;
    }

    double GPA = totalPoints / totalCredits;

    cout << "\nYour Semester GPA is: " << GPA << endl;

    return 0;
}
```

Sample Calculation (1st Semester Subjects):

Subject	Cr Hrs	Grade	Grade Pts	Quality Pts
Calculus & Analytical Geometry	3	A	4.0	12.0
Introduction to Computing	3	A-	3.7	11.1
Physics	3	B+	3.3	9.9
English Composition	3	B	3.0	9.0
Pakistan Studies	2	A	4.0	8.0
TOTAL	**14**			**50.0**
SEMESTER GPA				**50.0 / 14 = 3.57**